

Jakob Reinhardt, Dr.-Ing.

08.08.2025



+49 (0) 17661397646

jakob.reinhardt@gmail.com



I specialize in data analytics, human factors, and user research, with a PhD in human-robot interaction. I have professional experience in academia and the medical device industry, currently integrating robotics into clinical workflows. I mostly code in Python. I volunteer as a mountain rescuer and enjoy mountain sports with my family.

PROFESSIONAL

As of 05/2025 Postdoc Researcher, Chair of Ergonomics, TUM

- Research on Human Factors in Robotic Assisted Surgery
- Group Lead: Robotics For Life and Healthcare
- Speaker of the ForNeRo Project: <https://fornero.ed.tum.de/>
- Teaching: Seminar on Human-Robot Interaction

2021-2025 Data Analyst, Corpuls

- Developed the concept and data analysis for post-market clinical follow-up studies using medical field data from rescue missions.
- Released enterprise-wide Python tools for data analysis.
- Supported analytics software and data product development. Managed cross-department data requirements.
- Led inferential statistics workshops.
- Analyzed annotator behavior for dataset quality control.

2016-2021 Research Associate, Chair of Ergonomics, TUM

- Industry Projects
 - UNICARagil: Researched legible motion for robots and automated vehicles
 - IMAGinE: Designed human-machine interfaces for cooperative automated maneuver planning between cars;
 - Quick-Check Digitalization: Consulted medium sized businesses on Industry 4.0 applications
- Teaching
 - Taught the lab course on system ergonomics
 - Lectures on human factors of automated and cooperative driving
 - Supervised 20 bachelor and master theses
- Strategy
 - Conceptualized research and project proposals - acquired a DFG project: <https://gepris.dfg.de/gepris/projekt/453461364>;
 - Strategic advisory board of the institute establishing a strong link to MIRMI

ACADEMIC

2016-2021 PhD, Chair of Ergonomics, TUM

PHD THESIS

Parametrization and evaluation of legible motion for human-robot interaction

TUM - M.Sc. Mechanical Engineering (04/2013 - 09/2015)

RMIT Exertion Games Lab, Melbourne Australia (09/2013 - 12/2013)

TUM - B.Sc. Mechanical Engineering (10/2008 - 12/2012)

Aalto University, Helsinki Finland (01/2011 - 06/2011)



SOFTWARE & TOOLS

DATA ANALYSIS & SOFTWARE DEV.:

- Languages: Python, SQL, R
- Data handling: pandas, numpy etc.
- Visualization: plotly, holoviews
- Statistics: scipy, statsmodels
- Versioning & Automatization: Git, Gitlab CI/CD

COLLABORATION:

- Confluence, Jira, Teams, Git, Miro, VS Code, Jupyter, MS Office, LaTeX



KEY PUBLICATIONS



SOCIAL ENGAGEMENT

AS OF 2019 BERGWACHT BAYERN
Mountain rescue, Paramedic (SAN A/B)

UNTIL 2017 TECHNOLOGY WITHOUT BORDERS

Project lead for water supply in Ghana and photovoltaic system in Kenya



LANGUAGES

German - native
English - fluent



PERSONAL INTERESTS

Family: married, one daughter
Paragliding
Ski mountaineering
Coding projects on GitHub